



NCI Protocol #: 10323

Patient ID:

Date:

1 of 12

Patient Name:

Date Collected:

Biopsy Site: **Right upper lung**

Sample ID:

Telephone:

Tumor Content (%): **20_49**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Non-small cell lung cancer**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Relevant Non-Small Cell Lung Cancer Findings

Gene	Finding	Gene	Finding
ALK	None detected	NTRK1	None detected
BRAF	None detected	NTRK2	None detected
EGFR	EGFR D770_N771insG	NTRK3	None detected
ERBB2	None detected	RET	None detected
KRAS	None detected	ROS1	None detected
MET	None detected		

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	EGFR D770_N771insG	amivantamab ¹ mobocertinib ¹	None	40
III	TSC2 N314S	None	None	8

Public data sources included in relevant therapies: FDA¹, NCCN

Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Alerts informed by public data sources: ⛔ Contraindicated, 🛡️ Resistance

EGFR D770_N771insG 🛡️ afatinib, dacomitinib, erlotinib, gefitinib

Public data sources included in alerts: FDA¹, NCCN

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.06(006). The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Coverage	Transcript	Variant Effect
EGFR	p.(D770_N771insG)	c.2310_2311insGGT	COSM12378	chr7:55249012	20.92%	1989	NM_005228.3	nonframeshift Insertion
TSC2	p.(N314S)	c.941A>G	.	chr16:2108840	11.83%	947	NM_000548.3	missense

Comments:

The quality/quantity of RNA was insufficient to perform fusion analysis in this patient’s specimen. Repeat testing using a new biopsy/ FFPE block is recommended, if clinically desired.

Report Signed By

Name	Role	Date

Biomarker Descriptions

EGFR (epidermal growth factor receptor)

Background: The EGFR gene encodes the epidermal growth factor receptor (EGFR) tyrosine kinase, a member of the ERBB/human epidermal growth factor receptor (HER) family. In addition to EGFR/ERBB1/HER1, other members of the ERBB/HER family include ERBB2/HER2, ERBB3/HER3, and ERBB4/HER4¹. EGFR ligand induced dimerization results in kinase activation and leads to stimulation of oncogenic signaling pathways including the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways. Activation of these pathways promote cell proliferation, differentiation, and survival^{2,3}.

Alterations and prevalence: Recurrent somatic mutations in the tyrosine kinase domain (TKD) of EGFR are observed in approximately 10-20% of lung adenocarcinoma, and at higher frequencies in never-smoker, female, and Asian populations^{4,5,6,7}. The most common mutations occur near the ATP-binding pocket of the TKD and include short in-frame deletions in exon 19 (EGFR exon 19 deletion) and the L858R amino acid substitution in exon 21⁸. These mutations constitutively activate EGFR resulting in downstream signaling, and represent 80% of the EGFR mutations observed in lung cancer. A second group of less prevalent activating mutations include E709K, G719X, S768I, L861Q, and short in-frame insertion mutations in exon 20^{9,10,11,12}. EGFR activating mutations in lung cancer tend to be mutually exclusive to KRAS activating mutations¹³. In contrast, a different set of recurrent activating EGFR mutations in the extracellular domain include R108K, A289V and G598V and are primarily observed in glioblastoma^{8,14}. Amplification of EGFR is observed in several cancer types including 30% of glioblastoma, 12% of esophageal cancer, 10% of head and neck cancer, 5% of bladder cancer, and 5% of lung squamous cell carcinoma^{5,6,7,14,15}. Deletion of exons 2-7, encoding the extracellular domain of EGFR (EGFRvIII), results in overexpression of a ligand-independent constitutively active protein and is observed in approximately 30% of glioblastoma^{16,17,18}.

Potential relevance: Approved first-generation EGFR tyrosine kinase inhibitors (TKIs) include erlotinib¹⁹ (2004) and gefitinib²⁰ (2015), which block the activation of downstream signaling by reversible interaction with the ATP-binding site. Although initially approved for advanced lung cancer, the discovery that drug sensitivity was associated with exon 19 and exon 21 activating mutations allowed first-generation TKIs to become subsequently approved for front-line therapy in lung cancer tumors containing exon 19 or exon 21 activating mutations. Second-generation TKIs afatinib²¹ (2013) and dacomitinib²² (2018) bind EGFR and other ERBB/HER gene family members irreversibly and were subsequently approved. First- and second-generation TKIs afatinib, dacomitinib, erlotinib, and gefitinib are recommended for the treatment NSCLC harboring EGFR exon 19 insertions, exon 19 deletions, point mutations L861Q, L858R, S768I, and codon 719 mutations, whereas most EGFR exon 20 insertions, except p.A763_Y764insFQEA, confer resistance to the same therapies^{23,24,25,26}. However, in 2021, the irreversible tyrosine kinase inhibitor, mobocertinib²⁷ was FDA approved for the treatment of NSCLC with EGFR exon 20 insertion mutations. Additionally, in 2022, the FDA granted breakthrough therapy designation to the

Biomarker Descriptions (continued)

irreversible EGFR inhibitors, CLN-081 (TPC-064)²⁸ and sunvozertinib²⁹, for locally advanced or metastatic non-small cell lung cancer harboring EGFR exon 20 insertion mutations. In lung cancer containing EGFR exon 19 or 21 activating mutations, treatment with TKIs is eventually associated with the emergence of drug resistance³⁰. The primary resistance mutation that emerges following treatment with first-generation TKI is T790M, accounting for 50-60% of resistant cases⁸. Third generation TKIs were developed to maintain sensitivity in the presence of T790M. Osimertinib³¹ (2015) is an irreversible inhibitor indicated for metastatic EGFR T790M positive lung cancer and for the first-line treatment of metastatic NSCLC containing EGFR exon 19 deletions or exon 21 L858R mutations. Like first-generation TKIs, treatment with osimertinib is associated with acquired resistance. In this case, resistance is associated with the C797S mutation and occurs in 22-44% of cases³⁰. The T790M and C797S mutations may be each selected following sequential treatment with a first-generation TKI followed by a third-generation TKI or vice versa³². T790M and C797S can occur in either cis or trans allelic orientation³². If C797S is observed following progression after treatment with a third-generation TKI in the first-line setting, sensitivity may be retained to first-generation TKIs³². If C797S co-occurs in trans with T790M following sequential treatment with first- and third-generation TKIs, patients may exhibit sensitivity to combination first- and third-generation TKIs, but resistance to third-generation TKIs alone^{32,33}. However, C797S occurring in cis conformation with T790M, confers resistance to first- and third-generation TKIs³². Fourth-generation TKIs are in development to overcome acquired C797S and T790M resistance mutations after osimertinib treatment. EGFR targeting antibodies including cetuximab (2004), panitumumab (2006), and necitumumab (2016) are under investigation in combination with EGFR-targeting TKIs for efficacy against EGFR mutations. The bispecific antibody, amivantamab³⁴, targeting EGFR and MET was approved (2021) NSCLC tumors harboring EGFR exon 20 insertion mutations. The Oncoprex immunogene therapy quaratusugene ozeplasmid³⁵ in combination with osimertinib received a fast track designation from the FDA (2020) for NSCLC tumors harboring EGFR mutations that progressed on osimertinib alone. BDTX-189³⁶ was granted a fast track designation (2020) for the treatment of solid tumors harboring an EGFR exon 20 insertion mutation.

TSC2 (TSC complex subunit 2)

Background: The TSC2 gene encodes the tuberlin protein. TSC2 and TSC1 (also known as hamartin) form a complex through their respective coiled-coil domains³⁷. The TSC1-TSC2 complex is a negative regulator of the mTOR signaling pathway that regulates cell growth, cell proliferation, and protein and lipid synthesis³⁸. Specifically, the TSC1-TSC2 complex acts as a GTPase activating (GAP) protein that inhibits the G-protein RHEB and keeps it in an inactivated state (RHEB-GDP). GTP bound RHEB (RHEB-GTP) is required to activate the mTOR complex 1 (mTORC1). TSC1 and TSC2 are tumor suppressor genes. Loss of function mutations in TSC1 and TSC2 lead to dysregulation of the mTOR pathway^{37,39}. Inactivating germline mutations in TSC1 and TSC2 are associated with tuberous sclerosis complex (TSC), an autosomal dominant neurocutaneous and progressive disorder that presents with multiple benign tumors in different organs³⁷.

Alterations and prevalence: Somatic mutations are observed in up to 8% of skin cutaneous melanoma, 7% of uterine corpus endometrial carcinoma, and 4% of cervical squamous cell carcinoma^{6,7}.

Potential relevance: Currently, no therapies are approved for TSC2 aberrations.

Relevant Therapy Summary

☒ In this cancer type ☐ In other cancer type ☒ In this cancer type and other cancer types ☒ No evidence

EGFR D770_N771insG			
Relevant Therapy	FDA	NCCN	Clinical Trials*
mobocertinib	☒	☒	☒ (II)
amivantamab	☒	☒	☒
bevacizumab, osimertinib	☒	☒	☒ (III)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type

☐ In other cancer type

☒ In this cancer type and other cancer types

☒ No evidence

EGFR D770_N771insG (continued)			
Relevant Therapy	FDA	NCCN	Clinical Trials*
furmonertinib	×	×	● (III)
toripalimab, chemotherapy	×	×	● (III)
atezolizumab, bevacizumab, chemotherapy	×	×	● (II/III)
almonertinib	×	×	● (II)
amivantamab, chemotherapy	×	×	● (II)
amivantamab, lazertinib	×	×	● (II)
BEBT-109	×	×	● (II)
catequentinib, sintilimab	×	×	● (II)
ensartinib	×	×	● (II)
JMT-101, osimertinib	×	×	● (II)
osimertinib	×	×	● (II)
sintilimab	×	×	● (II)
sunvozertinib	×	×	● (II)
AP-L1898	×	×	● (I/II)
BLU-451	×	×	● (I/II)
FWD-1509	×	×	● (I/II)
GB263T	×	×	● (I/II)
HS-10376	×	×	● (I/II)
JFAN-1001	×	×	● (I/II)
MCLA-129	×	×	● (I/II)
ORIC-114	×	×	● (I/II)
YK-029A	×	×	● (I/II)
zipalertinib	×	×	● (I/II)
BAY-2927088	×	×	● (I)
ET0038	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type

☐ In other cancer type

☒ In this cancer type and other cancer types

☒ No evidence

EGFR D770_N771insG (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
lazertinib, amivantamab	×	×	● (I)
NIP-142 (China Resources Pharmaceutical)	×	×	● (I)
NX-019	×	×	● (I)
palcitoclax, osimertinib	×	×	● (I)
PLB-1004	×	×	● (I)
SPH3348, osimertinib	×	×	● (I)

TSC2 N314S

Relevant Therapy	FDA	NCCN	Clinical Trials*
atezolizumab + ipatasertib, inavolisib	×	×	● (II)
everolimus	×	×	● (II)
nab-rapamycin (Abraxis)	×	×	● (II)
samotolisib	×	×	● (II)
temsirolimus	×	×	● (II)
ipatasertib, atezolizumab	×	×	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

EGFR D770_N771insG

NCT ID	Title	Phase
NCT05607550	A Global, Phase III, Randomized, Multicenter, Open-Label Study to Investigate the Efficacy and Safety of Furmonertinib Compared to Platinum-Based Chemotherapy as First-Line Treatment for Patients With Locally Advanced or Metastatic Non-Small Cell Lung Cancer With Epidermal Growth Factor Receptor Exon 20 Insertion Mutations	III
NCT04785742	An Open, Multicenter Phase II Clinical Study to Evaluate the Safety and Efficacy of Almonertinib in Patients With Advanced NSCLC With Rare Mutations in EGFR	II

Clinical Trials Summary (continued)

EGFR D770_N771insG (continued)

NCT ID	Title	Phase
NCT05498428	A Phase II, Open-Label, Parallel Cohort Study of Subcutaneous Amivantamab in Multiple Regimens in Patients With Advanced or Metastatic Solid Tumors Including EGFR-mutated Non-Small Cell Lung Cancer	II
NCT04965090	A Phase II Single-Arm Study of Amivantamab (JNJ-61186372) and Lazertinib in Metastatic EGFR-mutant Lung Cancer With Progressive or New CNS Metastases on Previous Treatment	II
No NCT ID	A Multicenter, Open-Label Phase II Clinical Trial Of The Efficacy And Safety Of BEBT-109 In Patients With Locally Advanced Or Metastatic Non-Small Cell Lung Cancer With EGFR Exon 20 Insertion Mutations	II
NCT04974879	Osimertinib Combined With Bevacizumab in the Treatment Epidermal Growth Factor Receptor (EGFR) Exon 20 Insertions Local Advanced or Metastatic Non-Small Cell Lung Cancer	II
NCT03574402	An Open-label, Multi-center, Phase II Umbrella Study to Assess Efficacy of Targeted Therapy or Immunotherapy Directed by Next Generation Sequencing (NGS) in Chinese Patients With Advanced NSCLC (TRUMP)	II
NCT05466149	A Phase II, Multicenter, Open-Label Study to Assess the Efficacy and Safety of Furmonertinib in Patients With Locally Advanced or Metastatic Non-Small Cell Lung Cancer (NSCLC) With Epidermal Growth Factor Receptor (EGFR) Exon 20 Insertion Mutations	II
NCT05132777	A Phase II, Open Label, Multi-center Study to Assess the Efficacy and Safety of JMT101 Combined With Osimertinib in Patients With Locally Advanced or Metastatic Non-Small Cell Lung Cancer Harboring EGFR Exon 20 Insertion Mutations	II
No NCT ID	A Phase II Randomized Study of AP32788 in Advanced EGFR Exon 20 Mutant NSCLC Patients.	II
NCT03191149	Phase II Study of AZD9291 (Osimertinib) in Advanced NSCLC Patients With Exon 20 Insertion Mutations in EGFR	II
No NCT ID	A Prospective, Single-Center, Single-Arm Phase II Clinical Study For Advanced Non-Small Cell Lung Cancer With EGFR/HER2 Gene Exon 20 Insertion Mutations Treated with Sintilimab	II
NCT05559645	DZD9008 in Patients With Locally Advanced or Metastatic Non-Small Cell Lung Cancer (NSCLC) With EGFR Mutations: Cohort Study	II
NCT04993391	A Dose-escalation, Dose-extension and Efficacy-extension, Phase I/II Clinical Study to Evaluate the Safety, Tolerability, Pharmacokinetic Profile and Preliminary Efficacy of AP-L1898 Capsule for the Treatment of the Patients With Locally Advanced or Metastatic Non-small Cell Lung Cancer	I/II
NCT05241873	Phase I/II Study of BLU-451 in Advanced Cancers With EGFR Exon 20 Insertion Mutations	I/II
NCT05068024	A Phase I/II, Open-Label Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Anti-Tumor Activity of FWD1509 MsOH in Advanced Non-Small Cell Lung Cancer	I/II
NCT05332574	A Phase I/II, First-in-Human, Open-Label, Multicenter Study Evaluating the Safety, Tolerability, Pharmacokinetics and Efficacy of a Trispecific EGFR/cMET/cMET Antibody GB263T in Subjects With Advanced Non-Small Cell Lung Cancer (NSCLC) and Other Solid Tumors	I/II
NCT05435274	Phase I/II Clinical Study on the Safety, Tolerability, Pharmacokinetics and Effectiveness of HS-10376 in Patients with Advanced Non-small Cell Lung Cancer	I/II

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Clinical Trials Summary (continued)

EGFR D770_N771insG (continued)

NCT ID	Title	Phase
No NCT ID	A single/multicenter, open phase I/II clinical trial to evaluate the safety, tolerability, pharmacokinetics and antitumor activity of JFAN-1001 capsules in the treatment of patients with locally advanced or metastatic non-squamous non-small cell lung cancer	I/II
NCT04868877	Phase I/II Dose Escalation and Expansion Study Evaluating MCLA-129, a Human Anti-EGFR and Anti-c-MET Bispecific Antibody, in Patients With Advanced NSCLC and Other Solid Tumors	I/II
NCT04930432	A Phase I/II Study of MCLA-129, a Human Anti-EGFR and Anti-c-Met Bispecific Antibody, in Patients With Advanced NSCLC and Other Solid Tumors, Evaluating Safety, Pharmacokinetic Characteristics and Antitumor Activity	I/II
NCT02716116	A Phase I/II Study of the Safety, Pharmacokinetics, and Anti-Tumor Activity of the Oral EGFR/HER2 Inhibitor TAK-788 (AP32788) in Non-Small Cell Lung Cancer.	I/II
NCT03974022	A Phase I/II, Open-Label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics and Anti-tumor Efficacy of DZD9008 in Patients With Advanced Non-Small Cell Lung Cancer (NSCLC) with EGFR or HER2 Mutation	I/II
NCT05767866	Evaluation Of The Phase I Clinical Trial Of Safety, Tolerability, Pharmacokinetic Characteristics, And Preliminary Efficacy Of YK-029A Tablets In Patients With Advanced Non-Small Cell Lung Cancer (NSCLC)	I/II
NCT04036682	A Phase I/II, Open-Label, Multi-Center Trial to Assess Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Efficacy of CLN-081 in Patients With Locally-Advanced or Metastatic Non-Small Cell Lung Cancer Harboring EGFR Exon 20 Insertion Mutations Who Have Previously Received Platinum-Based Systemic Chemotherapy	I/II
NCT05099172	An Open Label, First-in-human Study of BAY 2927088 in Participants With Advanced Non-small Cell Lung Cancer (NSCLC) Harboring an EGFR and/or HER2 Mutation	I
NCT04858958	A Phase Ib, Randomized, Open-label, Multi-center Study to Evaluate the Preliminary Efficacy and Safety of Furmonertinib Mesylate in Patients With Locally Advanced or Metastatic Non-small Cell Lung Cancer Harboring EGFR Exon 20 Insertion Mutation	I
NCT05364073	A Phase Ib Dose Escalation and Dose Expansion Study Evaluating the Safety, Pharmacokinetics, and Antitumor Activity of Furmonertinib in Patients With Advanced or Metastatic (Stage IV) Non-Small Cell Lung Cancer (NSCLC) With Activating Epidermal Growth Factor Receptor (EGFR) or Human Epidermal Growth Factor Receptor 2 (HER2) Mutations	I
NCT04077463	An Open-label Phase I/Ib Study to Evaluate the Safety and Pharmacokinetics of JNJ-73841937 (Lazertinib), a Third Generation EGFR-TKI, as Monotherapy or in Combinations With JNJ-61186372, a Human Bispecific EGFR and cMet Antibody in Participants With Advanced Non-Small Cell Lung Cancer	I
No NCT ID	A single arm, open-label, multicenter phase Ia/Ib clinical trial to evaluate the safety, tolerance, pharmacokinetics and efficacy of NIP 142 capsule in patients with locally advanced / metastatic non-small cell lung cancer with EGFR mutation positive	I
NCT05514496	A First-in-Human, Open-Label, Dose Escalation and Expansion Study of Orally Administered NX-019 in Patients With Advanced, EGFR Mutant Cancer	I
NCT04001777	A Phase Ib Study of Safety and Efficacy of APG-1252 in Combination With Osimertinib (AZD9291) in EGFR TKI Resistant NSCLC Patients	I

Clinical Trials Summary (continued)

EGFR D770_N771insG (continued)

NCT ID	Title	Phase
NCT05347628	A Phase I, Open-label, Multicenter, Dose Escalation and Dose Expansion Study to Assess the Safety, Tolerability, Pharmacokinetics and Anti-tumor Efficacy of PLB1004 in Patients With Advanced Non-Small Cell Lung Cancer (NSCLC)	I
NCT05104281	Osimertinib Combined With Bevacizumab in Patients With Brain Metastasis Epidermal Growth Factor Receptor (EGFR) Mutation Positive Metastatic Non-Small Cell Lung Cancer	III
NCT03924050	A Randomized, Double-Blind, Placebo-Controlled, Multicenter, Phase III Study of Pemetrexed + Platinum Chemotherapy With or Without Toripalimab (JS001) in Advanced Non-small Cell Lung Cancer (NSCLC) Participants With TKI-resistant EGFR-mutated Tumors	III
NCT03178552	A Phase II/III Multicenter Study Evaluating the Efficacy and Safety of Multiple Targeted Therapies as Treatments for Patients With Advanced or Metastatic Non-Small Cell Lung Cancer (NSCLC) Harboring Actionable Somatic Mutations Detected in Blood (B-FAST: Blood-First Assay Screening Trial)	II/III
NCT03765775	A Randomized, Phase II Study of Anlotinib Combined With Sintilimab (IBI 308) in First-generation EGFR-TKIs Drug Resistance Along With T790M Negative NSCLC	II
NCT05088070	Phase I Clinical Study of SPH3348 Tablets, a c-Met Inhibitor, in Patients With Advanced Solid Tumors With c-Met Abnormalities	I
NCT05315700	A Phase Ib/II, Single agent, Tumor-Agnostic Trial of ORIC-114 in Patients With Advanced Solid Tumors With EGFR or HER2 Exon 20 Alterations or HER2 Amplification And Will allow Patients With CNS Metastases that are Either Treated or Untreated But Asymptomatic.	I/II
NCT05354843	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of SHP2 Inhibitor ET0038 Monotherapy in Patients With Advanced Solid Tumors	I

TSC2 N314S

NCT ID	Title	Phase
NCT04185831	MEGALiT - a Multicenter, Basket and Umbrella Explorative Trial on the Efficacy and Safety of Molecular Profile Selected Commercially Available Targeted Anti-cancer Drugs in Patients With Advanced Cancers Progressive on Standard Therapy	II
NCT03155620	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) Screening Protocol	II
NCT03213678	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase II Subprotocol of LY3023414 in Patients With Solid Tumors	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT05103358	A Phase II Multi-center Open-label Basket Trial of ABI-009 (Nab Sirolimus) for Adult and Adolescent Patients With Solid Tumors Harboring Pathogenic Inactivating Alterations in TSC1 or TSC2 Genes	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II

Clinical Trials Summary (continued)

TSC2 N314S (continued)

NCT ID	Title	Phase
NCT03673787	Ice-CAP: A Phase I Trial of Ipatasertib in Combination With Atezolizumab in Patients With Advanced Solid Tumours With PI3K Pathway Hyperactivation.	I/II

Assay Information

Methodology, Scope, and Application:The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity:The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations:This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient’s history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient’s) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content:Thermo Fisher Scientific’s Ion Torrent Oncomine Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of Oncomine Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in ‘Other Identifiers.’ Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:
AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSP02, RSP03, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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